Global Hurricane Catastrophe Loss Model

Project Summary

Section 1: Overview

This project constructs a multi-basin Monte Carlo catastrophe model that simulates annual hurricane wind intensity across five global ocean basins. By modeling both storm frequency and severity using established actuarial distributions, the model produces a full distribution of annual loss outcomes — enabling the calculation of key risk metrics used in the insurance and reinsurance industries. This type of model is used by insurers, reinsurers, and real estate firms to determine how much capital should be held in reserve against catastrophic storm events.

Section 2: Data Source

Storm data was sourced from the International Best Track Archive for Climate Stewardship (IBTrACS), maintained by NOAA's National Centers for Environmental Information (NCEI). The dataset spans 1980 to 2025 and, after filtering, contains 2,197 hurricane-strength storms across five ocean basins. Only storms with maximum sustained winds of 64 knots or greater were included, consistent with the World Meteorological Organization's threshold for hurricane classification. The South Atlantic basin was excluded from modeling as only a single qualifying storm was recorded over the entire period, making reliable parameter estimation impossible.

Section 3: Methodology

Storm frequency per year was modeled using a Poisson distribution, which is the standard actuarial choice for counting rare, independent events over a fixed time period. Storm intensity was modeled using a lognormal distribution, reflecting the right-skewed nature of hurricane wind speeds — most storms are moderate, but occasional extreme events produce disproportionately high intensity values.

Physical wind speed (knots) was used as the severity metric rather than dollar losses. This decision reflects two key constraints: first, insured dollar losses vary significantly by country based on wealth and insurance penetration, making global comparisons unreliable; second, consistent global loss data does not exist for the full study period, whereas IBTrACS provides standardized wind measurements across all basins.

Rather than fitting a single global model, five independent basin-level models were constructed — one each for the Western Pacific (WP), Eastern Pacific (EP), South Indian (SI), South Pacific (SP), and North Indian (NI) basins. This stratified approach reflects the materially different storm climatology across basins, each of which has distinct frequency and intensity characteristics. Each basin's lambda (λ), mu (μ), and sigma (σ) parameters were derived directly from historical IBTrACS data. Ten thousand years were then simulated per basin using Monte Carlo methods.

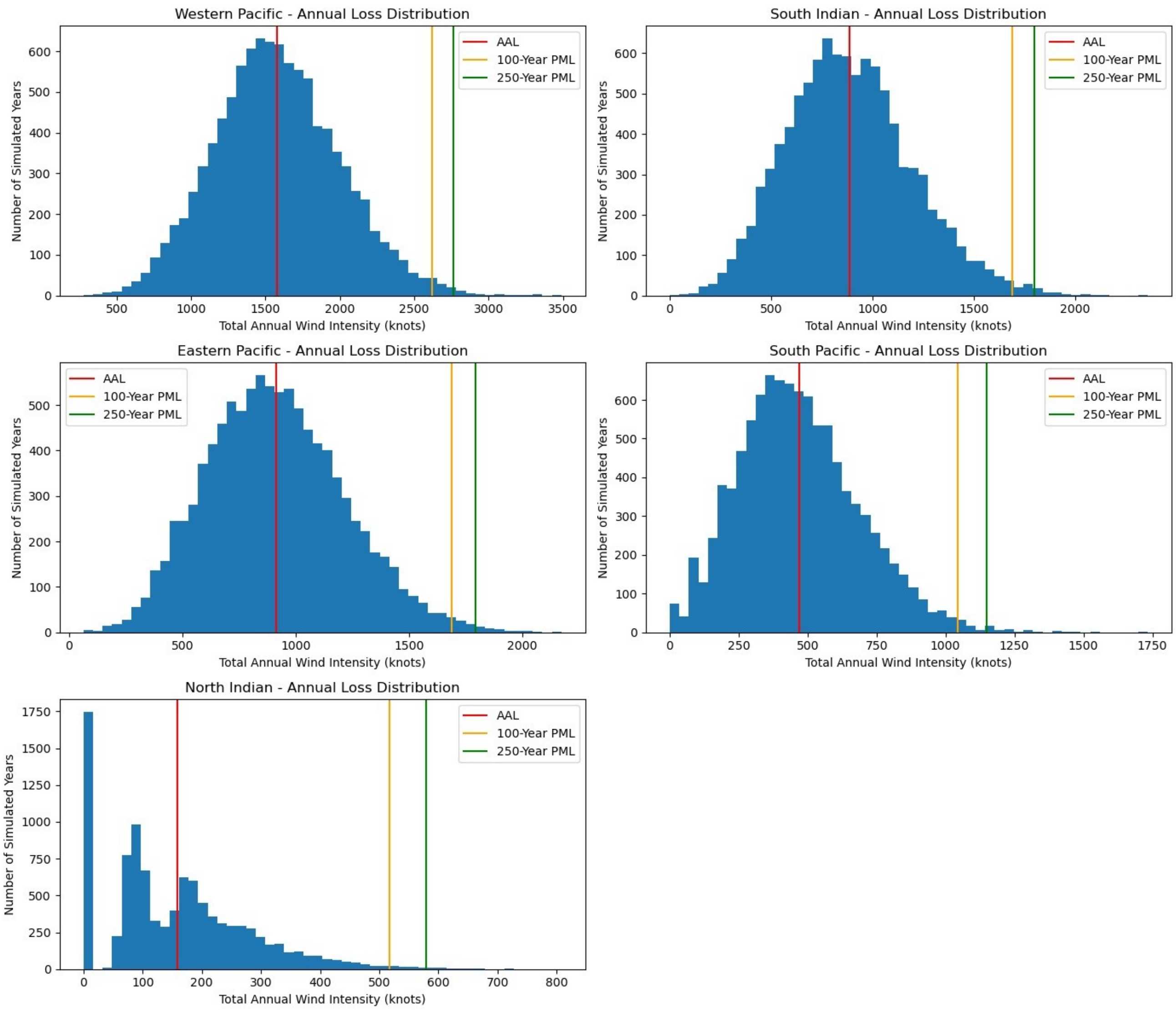
Section 4: Results

The table below presents the Average Annual Loss (AAL) and Probable Maximum Loss (PML) at the 100-year and 250-year return periods for each basin. All values are expressed in accumulated annual wind intensity (knots).

Basin	Full Name	AAL (knots)	100-Year PML	250-Year PML
WP	Western Pacific	1,588	2,663	2,859
EP	Eastern Pacific	907	1,703	1,816
SI	South Indian	889	1,674	1,780
SP	South Pacific	471	1,051	1,159
NI	North Indian	159	504	561

The Western Pacific dominates all other basins, with an AAL nearly ten times that of the North Indian — directly reflecting its substantially higher storm frequency ($\lambda = 15.3$ vs $\lambda = 1.7$). Notably, the Eastern Pacific and South Indian basins produce remarkably similar risk profiles despite being in different hemispheres, suggesting comparable storm climatology. The North Indian basin exhibits the most pronounced tail risk relative to its average: its 100-year PML is approximately 3.2 times its AAL, compared to 1.7 times for the Western Pacific, indicating that while North Indian storms are rare, extreme years are disproportionately severe.

The charts below show the full simulated loss distribution for each basin, with vertical lines marking the AAL (red), 100-year PML (orange), and 250-year PML (green):



Section 5: Limitations

The following limitations should be considered when interpreting model results:

- Wind speed as a severity proxy: Wind speed is a physical intensity measure, not a direct representation of insured losses. Actual financial losses depend on storm track, affected population, building vulnerability, and insurance penetration — none of which are captured in this model.
- Parameter uncertainty: While 2,197 storms provide a reasonable sample, 46 years of observation may underrepresent the true frequency of rare, extreme events. Return period estimates at the tail (250-year PML) carry greater uncertainty than near-average estimates.
- Climate stationarity assumption: The model assumes that historical storm parameters remain constant over time. In reality, climate change may be altering hurricane frequency and intensity, potentially making historical parameters a poor guide to future risk.

- Basin independence: Each basin is modeled independently, ignoring known climatic teleconnections. For example, El Niño conditions simultaneously suppress Atlantic storm activity while enhancing Eastern Pacific activity. A correlated multi-basin model would better reflect this dynamic.
- No landfall adjustment: The model counts all hurricane-strength storms regardless of whether they made landfall. Storms that remain over open ocean cause no insured loss. A more refined model would weight severity by proximity to populated coastlines.

Data Source: NOAA NCEI IBTrACS v04r01 (1980–2025) | Model built in Python using NumPy, SciPy, pandas, and Matplotlib